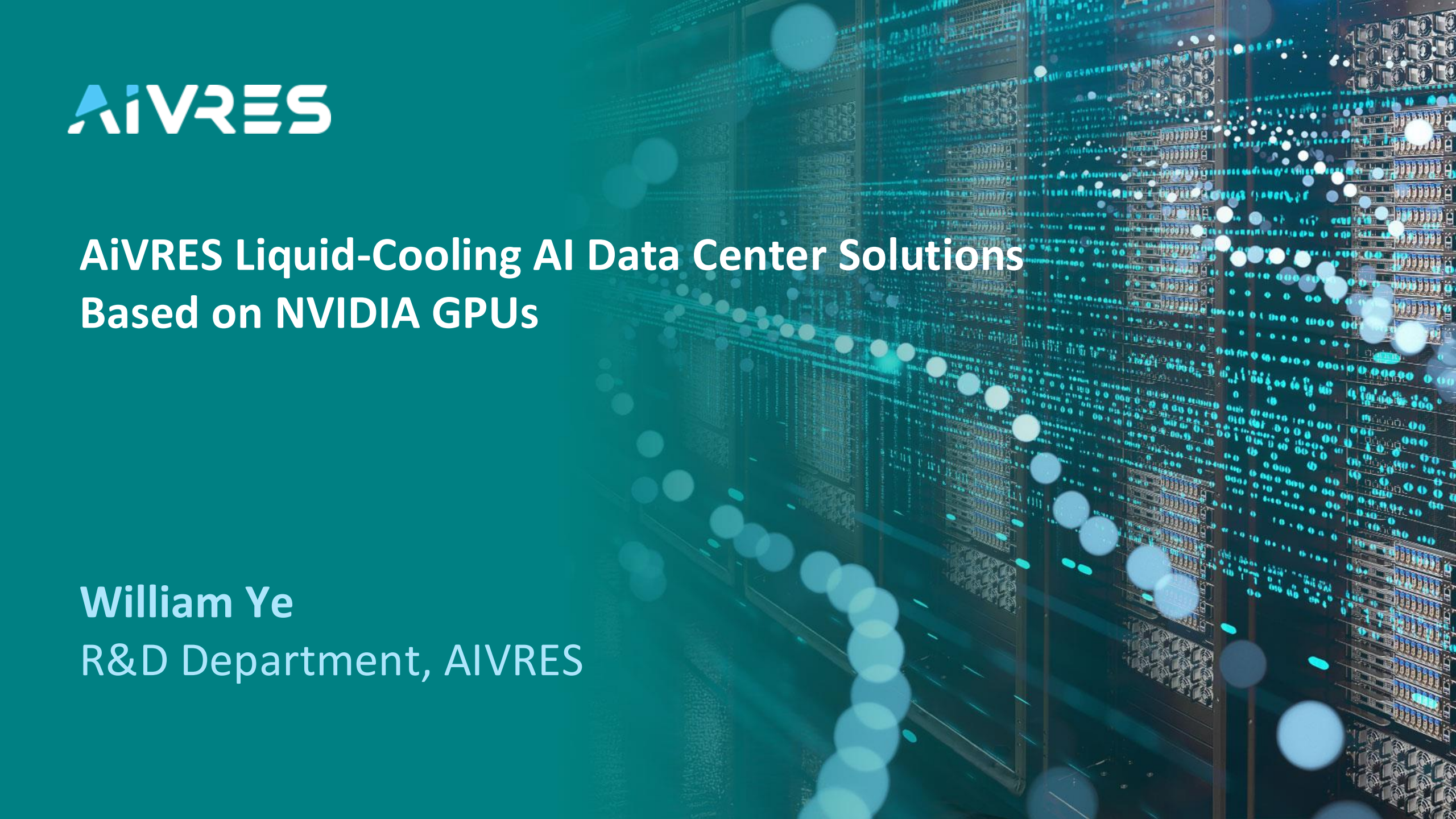


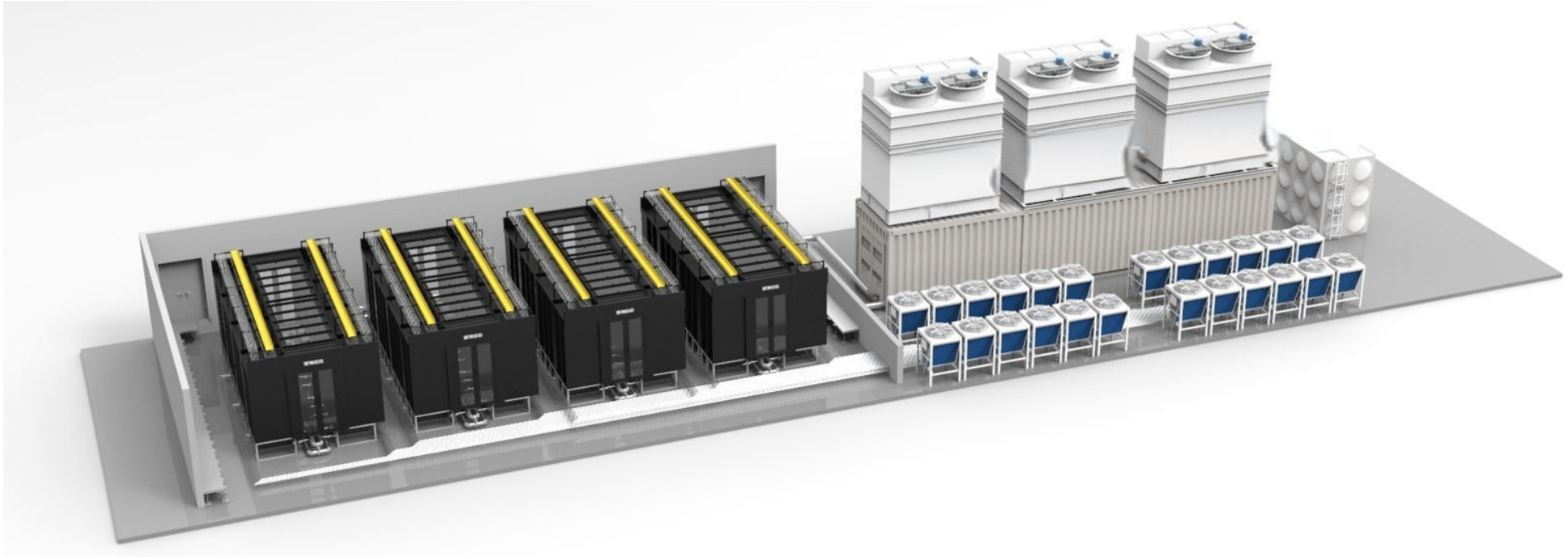
AiVRES Liquid-Cooling AI Data Center Solutions Based on NVIDIA GPUs

William Ye
R&D Department, AIVRES

1

Liquid Cooling AI Data Center Architecture

AI Data Center Architecture



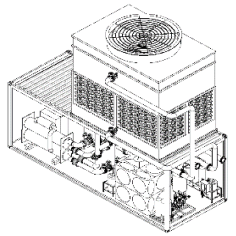
Cooling Source and Water Treatment Device



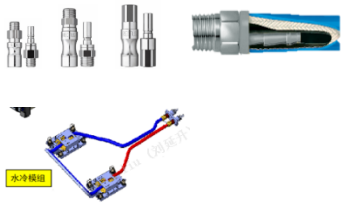
Open tower



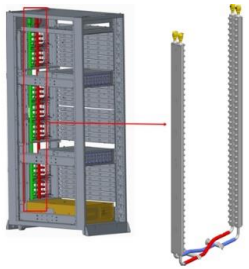
Cooling Machine



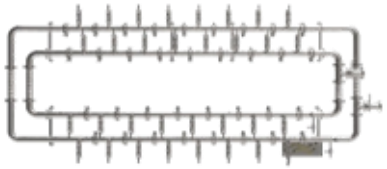
Integrated Cooling Source



Liquid Cooling components



Liquid Cooling Cabinet



Secondary Cooling Loop



CDU



Liquid - Air



Liquid-Liquid



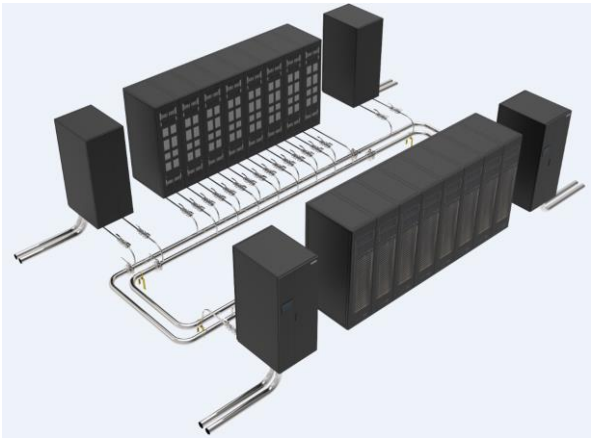
Micromodule (Closed Loop)



Air-Cooled Air Conditioner

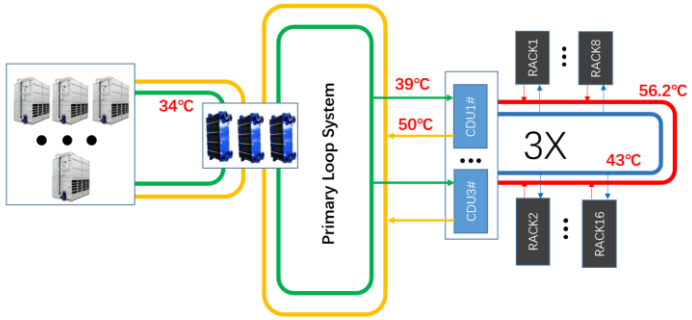
Row-Level Computing Power Modules: Large-Scale and Efficient Deployment

POD Level Deployment



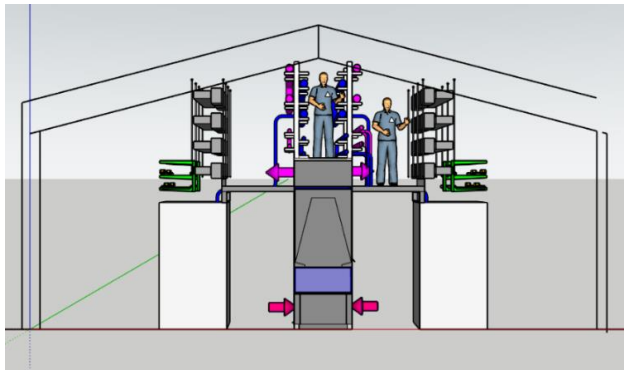
- Single-row/double-row POD design, single POD uses 16 cabinets for deployment.
- CDU uses 2N/N+1 deployment for high reliability.

Wide Geographical Deployment



- Uses high performance heat exchange megawatt-class CDU.
- CDU extreme performance mode solves the problem of having a high water temperature in tropical areas.

Safety and Reliability



- Upper pipe routing, two-layer design provides convenient maintenance.
- Uses water collection tray and leakage monitoring system. Supports pipeline leakage monitoring.
- Supports removal of the cabinet system when leakage occurs thus improving liquid cooling safety.

2 Innovative Liquid Cooling Products

Megawatt-Class CDU: Safe and Efficient Liquid Cooling



Megawatt-Class CDU

- High Efficiency: Using a Class 1 energy-efficient circulation pump, the overall efficiency ratio is close to 100.
- Safety: The primary and secondary loops are equipped with filters. An optional solution monitoring system can be configured.
- Compatibility: Utilizes a bypass system. This allows the CDU to adjust the liquid supply to accommodate the entire load range.
- Communication Capabilities: Supports secure communication and remote parameter configuration.

Rated Heat Exchange Note 1		kW	≥1200
Primary Side	Circulating Medium	/	Deionized water/PG25/EG25
	Maximum Temperature	℃	≤41
	Maximum Flow Rate	L/min	1400
Secondary Side	Circulating Medium	/	Deionized water/PG25
	Inlet and Outlet Temperature	℃	60/45
	Secondary Side Filtration	μm	50、25
	Differential Pressure Adjustment Range	bar	0.3bar~2bar
	Secondary Pump Flow Rate	L/min	≥1200
Electrical System	Power supply requirements	kW	≤15 kW
	Power Supply	/	AC380V~480 50hz/60hz
	Communication Ports	/	RS485/RJ45
	Communication Protocol	/	Modbus-RTU/ Modbus-TCP/SNMP
	Dimensions (width*depth*height)	mm	900*1200*2200

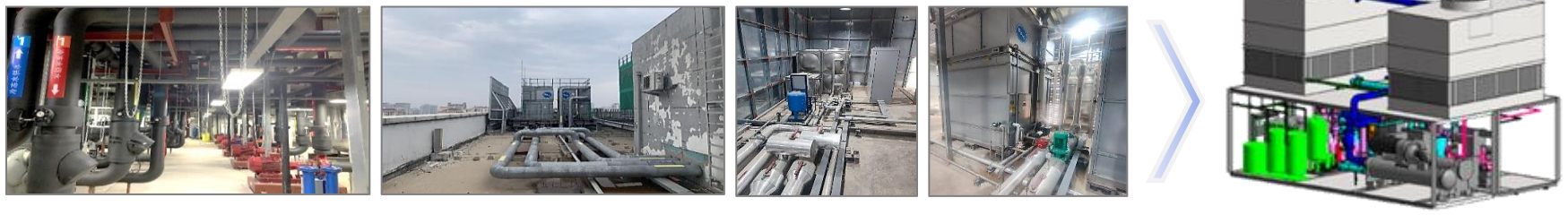
Note 1： Primary Side EG25, 41℃ inlet, 54℃ outlet; Secondary Side PG25, 45℃ supply, 65℃ return

Integrated Cold Source: Fast Primary Loop Deployment



Integrated Cold Source

- Efficient Delivery: Prefabricated construction, factory-based provisioning, and on-site "plug and play" can greatly shorten the construction period.
- Green / Energy Savings: Variable frequency water pumps and cooling towers, allows for decentralized independent control of each device. This ensures the highest operating efficiency of the overall cooling system.
- Safe and Reliable: Redundant configuration of essential equipment, including cooling towers and water pumps ensures reliability. Off-site prefabrication, inspection, and testing ensures product performance.
- Remote Capabilities: Design allows for unmanned operation, remote control of equipment, alert notifications, and fault diagnosis.



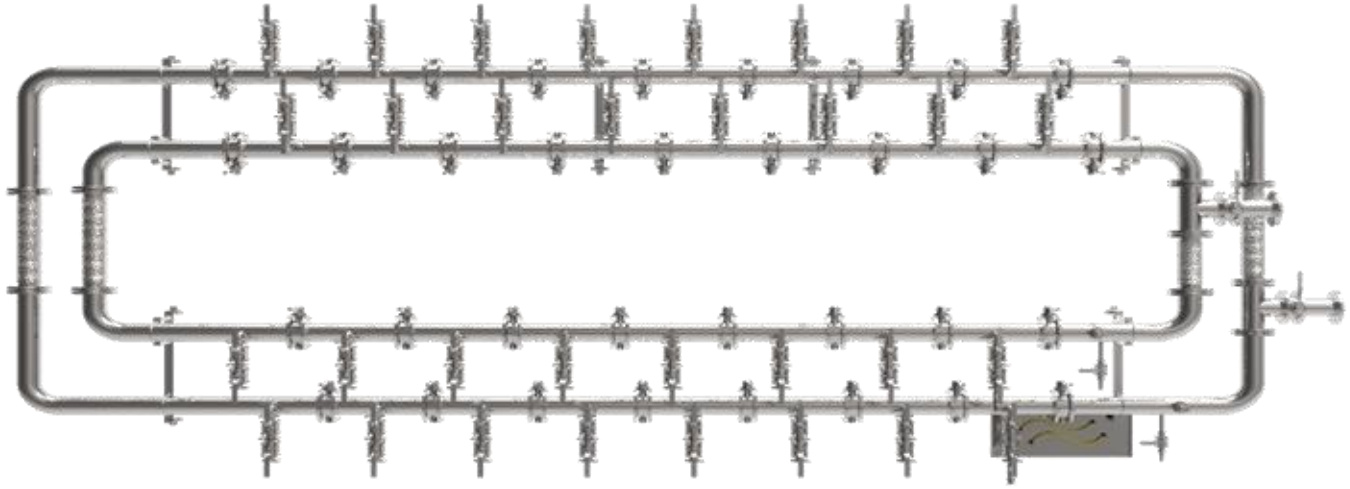
Secondary Cooling Loop: Quick Installation and Easy Maintenance

Standardized Installation

- Prefabricated standardized stainless-steel pipes avoids welding in the construction site.
- Use of a ring pipe network structure avoids a single point of failure. This ensures reliability, hydraulic balance, and uniform heat dissipation in each rack cabinet.

Easy Maintenance

- The dual valve design only requires a single rack to be disconnected when a single valve fails.
- The dual valve design between branches reduces the loss of coolant during cabinet maintenance.



Integrated Monitoring Platform

2D/3D Visualization

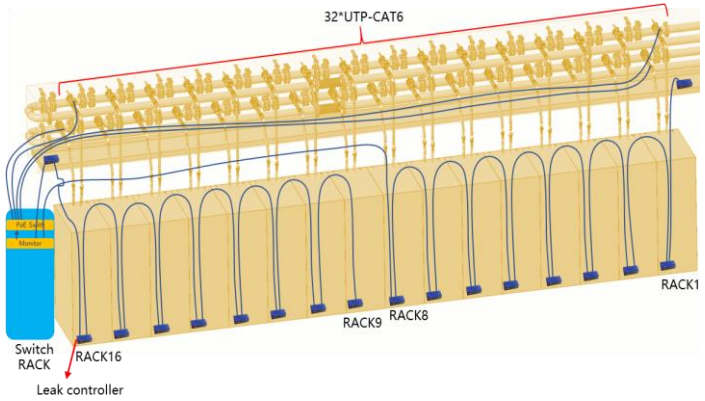
- 2D/3D display of the data center gives a comprehensive view of the facility. This provides a visual representation of all the equipment.
- 3D display of assets provides configuration information, accurate asset location, inventory of asset quantity, and real-time monitoring of equipment changes.



3D Layout Display

Precise Water Leakage Control

- The platform monitors for any leakages. Is it capable of closing a specific valves to any rack in the entire system.
- Design provides simple and convenient wiring and installation. All valve control and leakage monitoring requires just a wired network connection (RJ45). No additional power supply is required.



Leakage Control

Rack Level Flow Control

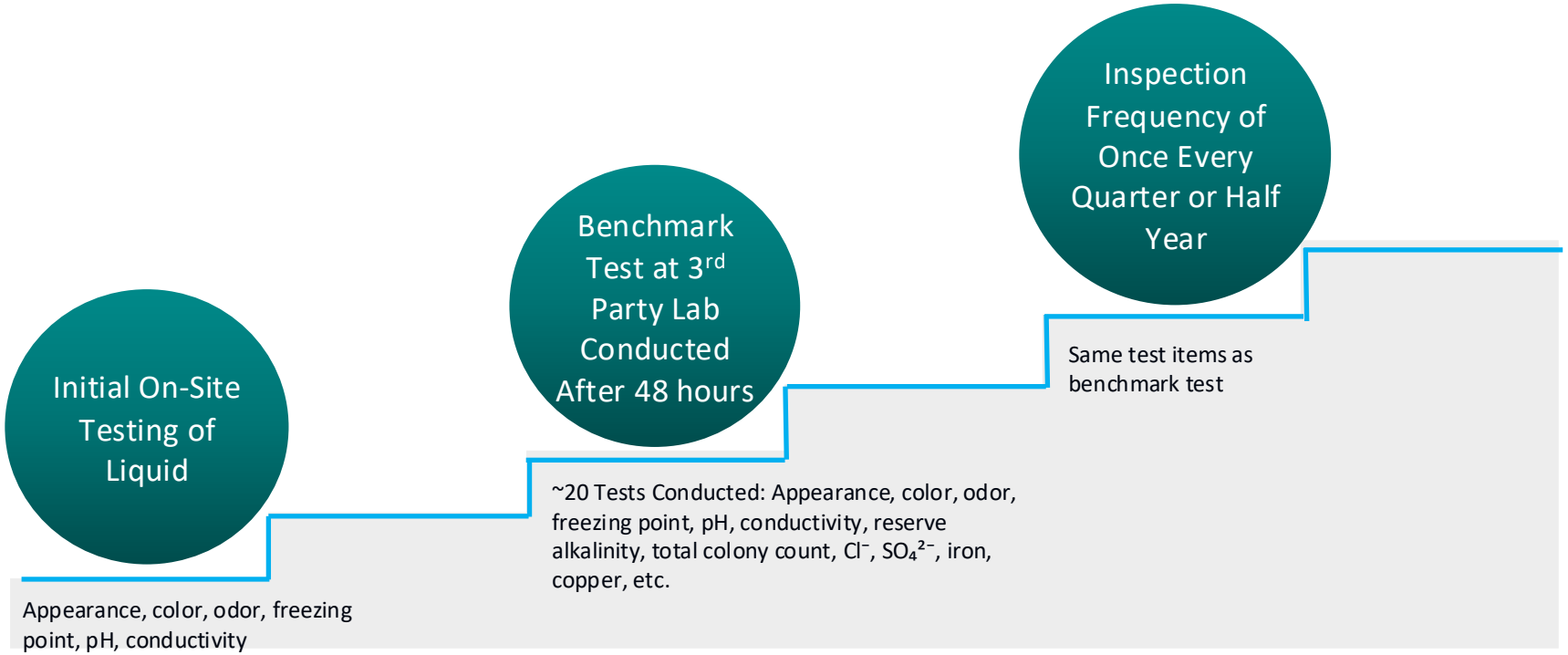
- The platform monitors each server's thermal and power consumption. Based off the information gathered, it can change the flow rate to each rack via the control valves. The flow rate can be reduced by up to 50%.



Valve Control

Operation and Maintenance Life Cycle

- Flushing Before Filling: Flush the pipe system with grade 3 deionized water before filling the coolant to avoid any contamination.
- Maintenance Inspection: Regularly sample and test the coolant during the operation. Send to a trusted third-party lab to provide an accurate, unbiased test report.
- Maintenance Requirements: Coolant must be replenished with the original brand and model of coolant.

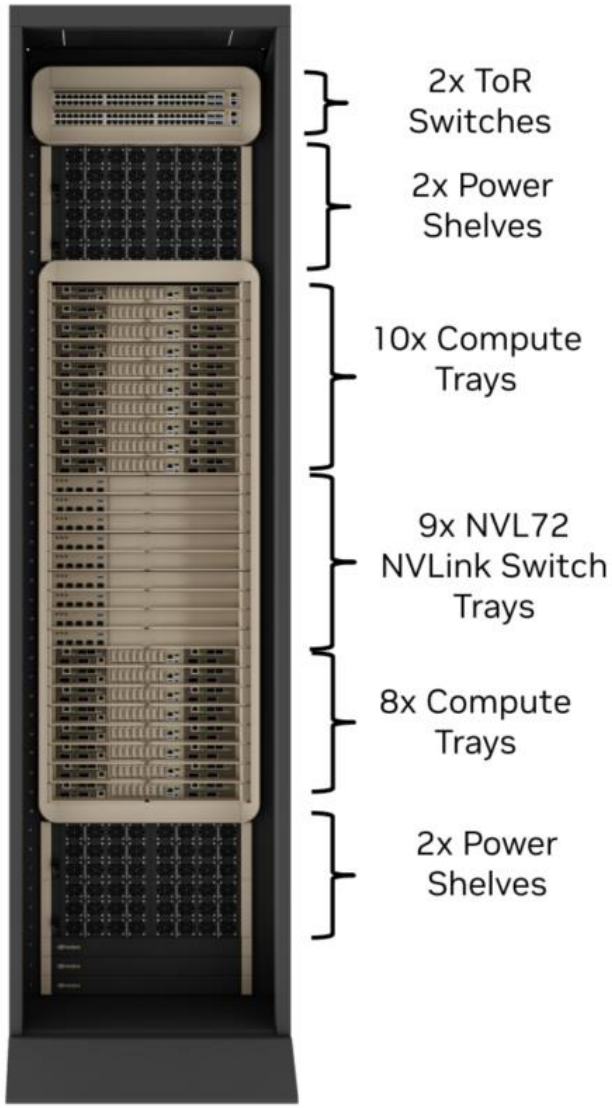


Coolant full life cycle testing and maintenance services

3

Liquid Cooled Aivres KRS8000V4 with NVIDIA VR NVL72, KL2288/KL3288 with NVIDIA HGX™ Rubin NVL8

KRS8000V4 VR NVL72 Rack for Real-Time Trillion LLM & MoE



Real-Time Trillion-Parameter LLM & MoE:

- Designed for interactive reasoning, long-context agents, and real-time serving of 1T+ LLM and MoE models.
- A 72-GPU NVLink 6 domain keeps high-volume GPU communication inside one scale-up fabric.

Graph Neural Networks:

- Accelerates memory- and communication-intensive GNN training on massive graph datasets.
- Ideal for recommendation, fraud detection, knowledge graphs, and scientific discovery.

Massive-Scale Model Training & Inference:

- Supports pretraining, post-training, test-time scaling, and production inference for frontier models.
- High compute density and scale-up bandwidth reduce cluster sprawl and accelerate time-to-model.

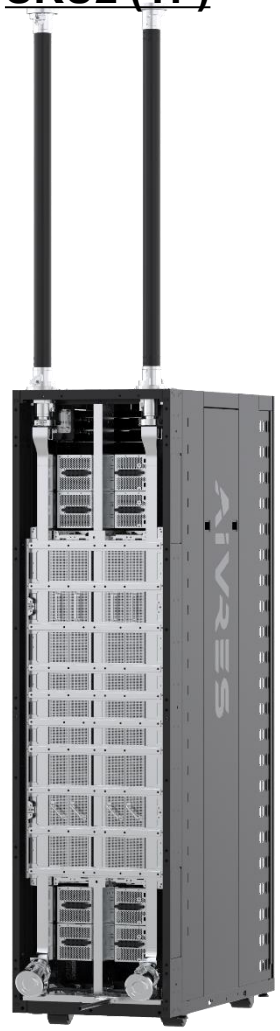
KRS8000V4 VR NVL72 Product Specification



SKU1 (BF)

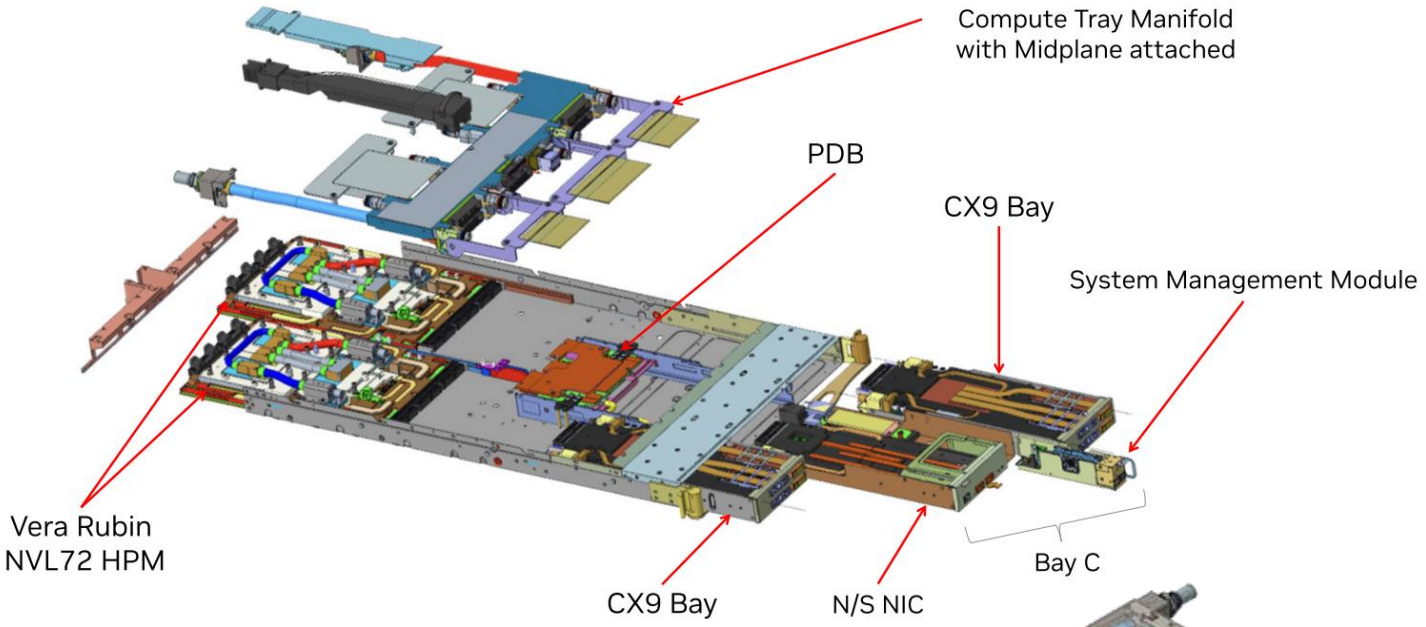


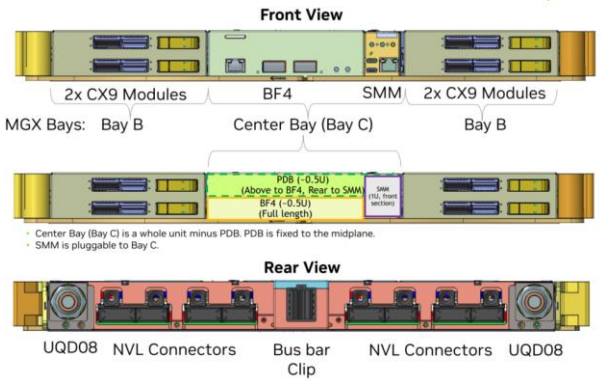
SKU2 (TF)

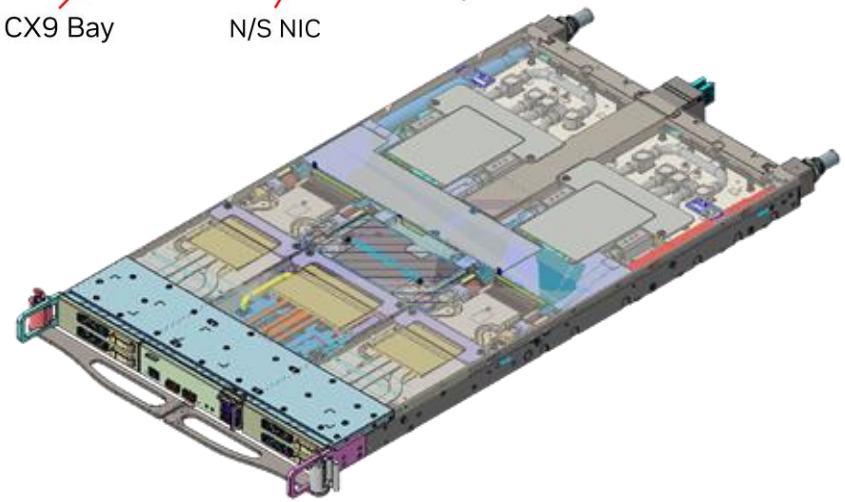


SKU Specification	KRS8000V4
Form Factor	48U
Processor	36x Nvidia **Vera**
GPU	72pkgx Nvidia Rubin, TDP **MaxQ 1800W / MaxP 2300W** (Q=Quality, P=Performance)
Memory	**27.648TB** SOCAMM (CPU) **20.736TB** HBM4 (GPU)
Drive Bays	**72**x E1.S NVMe SSD (Data) 18x **E1.S** NVMe SSD (Boot)
OCP	None
PCIe	18x PCIe**6.0** x16 with 1x **BF4** 18x PCIe**6.0** x16 Mezzanine Slots on board with **8x** 800G OSFP
PSU	**4x 3U 110kW** Power Shelf **2+2(MaxQ), 3+1(MaxP)**
Cooling Options	**96%** Liquid + **4%** Air Cooling: L2A Sidecar or L2L In-Row
Fan	**None**
Management	DC-SCM BMC Module
Security	TPM 2.0
Dimensions	600 mm (W) x **2300** mm (H) x 1200 mm (D)
Weight	**1,843.4 kg**
Power Consumption	TDP: **~187kW/~227kW**; EDP: **~198kVA/~240kVA**
Suitable Application	Ultra-Scale AI Training & Real-Time Inference for Trillion-Parameter LLMs, Big Data Analytics, and HPC Simulations

KRS8000V4 Compute Tray Design

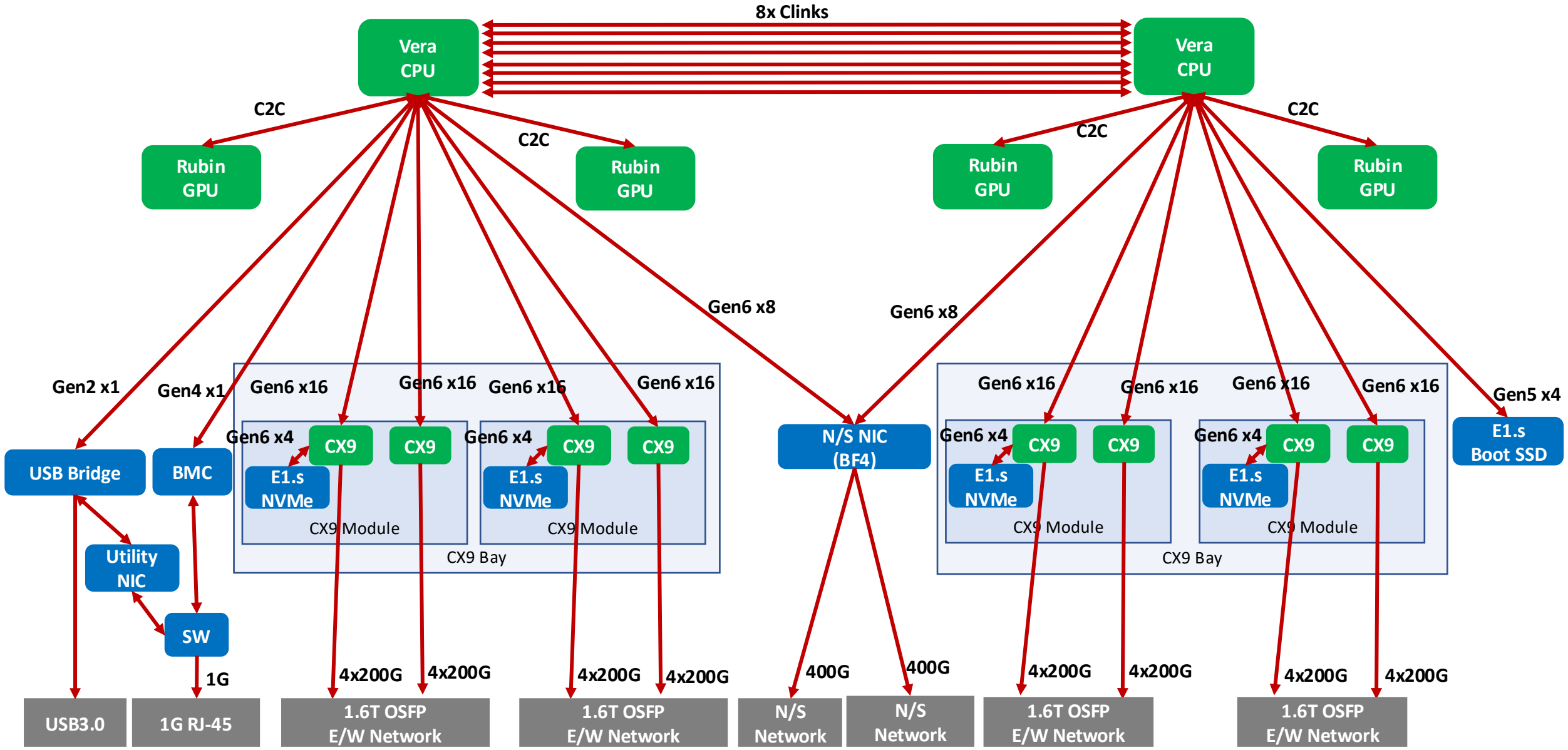






Compute Tray	
CPU/GPU	2xVera CPUs + 4xRubin GPUs (MaxQ: 1800W/MaxP: 2300W)
CPU Memory	1.44 TB SOCAMM
Cooling	1U LC 100%
Data Storage	4 x E1.S NVMe SSDs
Boot Storage	1 x E1.S NVMe SSDs
Front I/O	1 x USB 3.0, 1x Mgmt I/O, 1 x RJ45, 1 x Mini Display Port
N/S BF4	1 x BF4, 2 x 400G Ports
E/W NIC	4*CX9 - 8x OSFP Ports, with 1600GbE bandwidth to each GPU
Management	DC-SCM BMC Module
TPM	TPM2.0

KRS8000V4 Compute Tray Topology



E2E Datacenter Solution for KRS8000V4

AI INFRASTRUCTURE MANAGEMENT



MANAGEMENT CLUSTER



BCM/Kubernetes
KR2280N4

UFM
KR2280E3

MULTI-DEVICES MANAGEMENT



OOB NETWORK



MSN2201 1GbE/100GbE
N/S TOR MGM Network

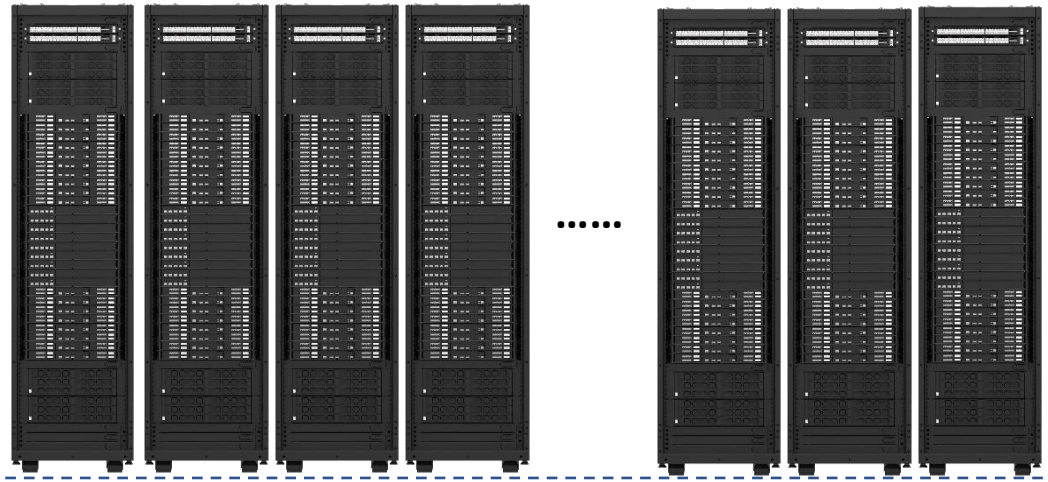
COMPUTING CLUSTER

Liquid CDU Liquid CDU



In Row CDU 3840kW In Row CDU 3840kW

KRS8000V4 Cluster – 16/14 racks per SU



COMPUTE FABRIC ALGORITHM & LLM

E/W high speed Network
Q3400 IB XDR/KS800 Eth



HIGH PERFORMANCE STORAGE CLUSTER





Flash Storage
KR1180E3
KR2190E3

VAST&WEKA Certify

Rubin HGX NVL8 Solutions Overview

SKU1: KL2288 2U
Full liquid compute tray
NV Reference Design



Deployment Scenarios:
Ideal for hyperscale AI farms that prioritize maximum GPU density and continuous high-power liquid cooling for sustained, high-throughput inference and training.

SKU2: KL3288 3U
Hybrid air-liquid cooled compute tray
More and flexible IO



Deployment Scenarios:
Best for customers needing flexible I/O and the largest NVLink domains—suited to multi-tenant inference platforms and long-context LLM services that require high inter-GPU bandwidth.

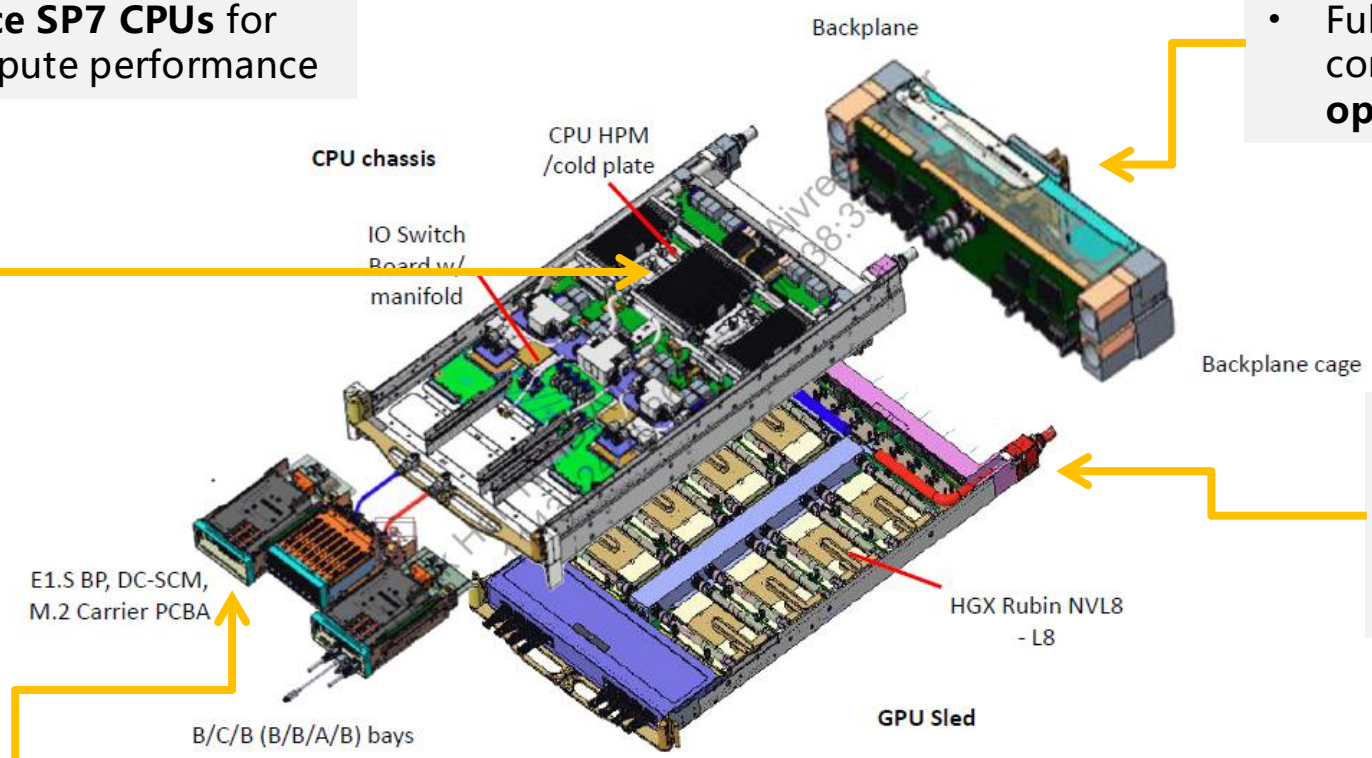
HGX Rubin Key Spec

	Whole Rack Solution	
	KL2288 (2U) 	**KL3288 (3U)** 
Form Factor	2U compute sled for rack	3U compute sled for rack
Processor	2x Intel ® Xeon® 6 processors GNR-SP (Solution reserved) 2x AMD EPYC 9006 Venice SP7	2x Intel ® Xeon® 6 processors GNR-**AP** (3U) 2x AMD EPYC 9006 Venice SP7 (Solution reserved)
Memory	24/32x DDR5-6400 RDIMMs, up to 4TB DDR5 ECC for Intel 32x DDR5-8000 RDIMMs, up to 4TB DDR5 ECC for AMD	
GPU	NVIDIA HGX Rubin 8-GPU	
Storage	8x NVMe E1.S + 2x NVMe M.2	Up to 10x 2.5" SAS/SATA/U.2/E3.S SSD + 2x NVMe M.2
PCIe	1x PCIe x16 FHHL for BF4	5x slots (3* LP AIC + 2* FHHL) for any NIC/storage 1x OCP slot (optional)
Network Interface	8x OSFP (IB XDR, ETH 800G) from CX-9	
Power	54V Rack Level Busbar	
Cooling	100% liquid cooling	Hybrid liquid and air cooling

HGX Rubin Highlights-KL2288

1. Next-Generation x86 Performance
- Dual **AMD Venice SP7 CPUs** for leading x86 compute performance

2. Lower OPEX & Better PUE
- Full liquid cooling reduces power consumption and **lowers long-term operating costs**



3. Sustained High Performance
- Superior thermal management enables stable, consistent performance under continuous AI workloads.

5. Proactive Hardware Root of Trust
- Active Real-Time Monitoring
- Comprehensive Firmware Resilience

4. Ease of Serviceability
- Front load M.2 design allows quick replacement without opening the cover, reducing MTTR

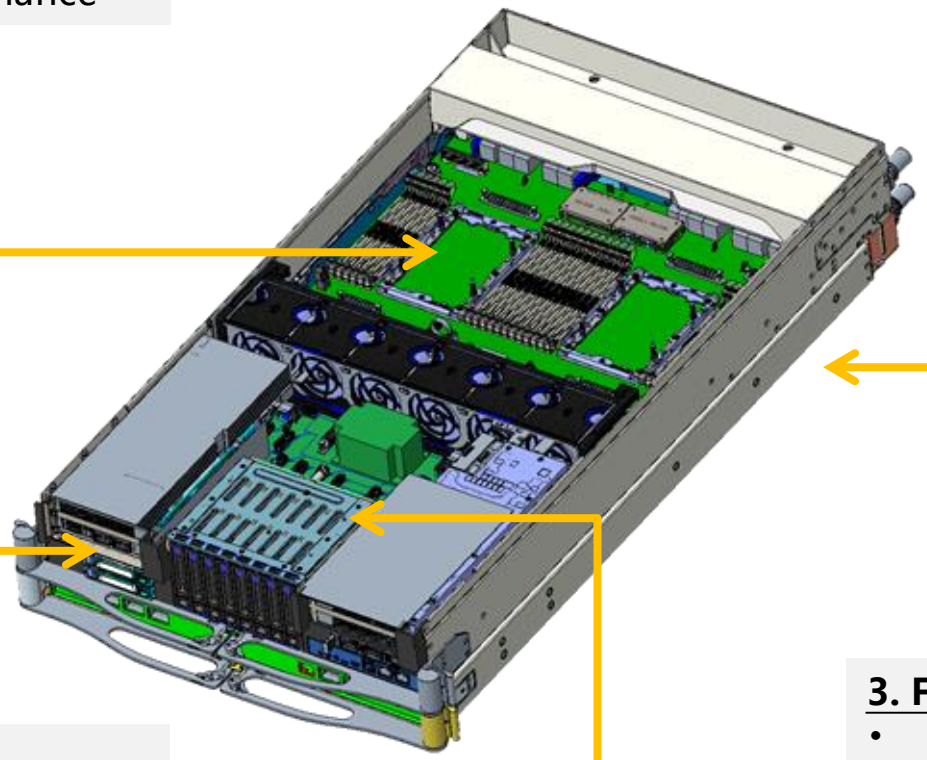
HGX Rubin Highlights-KL3288

1. Next-Generation x86 Performance

- Dual **Intel Xeon 6 GNR-AP CPUs** for leading x86 compute performance

2. Better CAPEX – Lower Initial Investment

- Hybrid cooling with **fewer cold plates and reused air-cooled components** reduces material and system build costs, lowering the initial investment.



3. Flexible I/O & Storage Options

- Support up to **10× U.2 drives and 4× PCIe cards**, providing flexible configurations for diverse **storage, management networking, and RAID requirements**, while improving material availability.

4. Easier Maintenance

Cold plates are dedicated to **CPU and GPU cooling**, while DIMMs, NICs, and storage remain air-cooled for **simpler access, serviceability, and maintenance**.

Rack Solutions for KL2288/3288

4 servers per rack (Lower Power, Higher Flexibility)



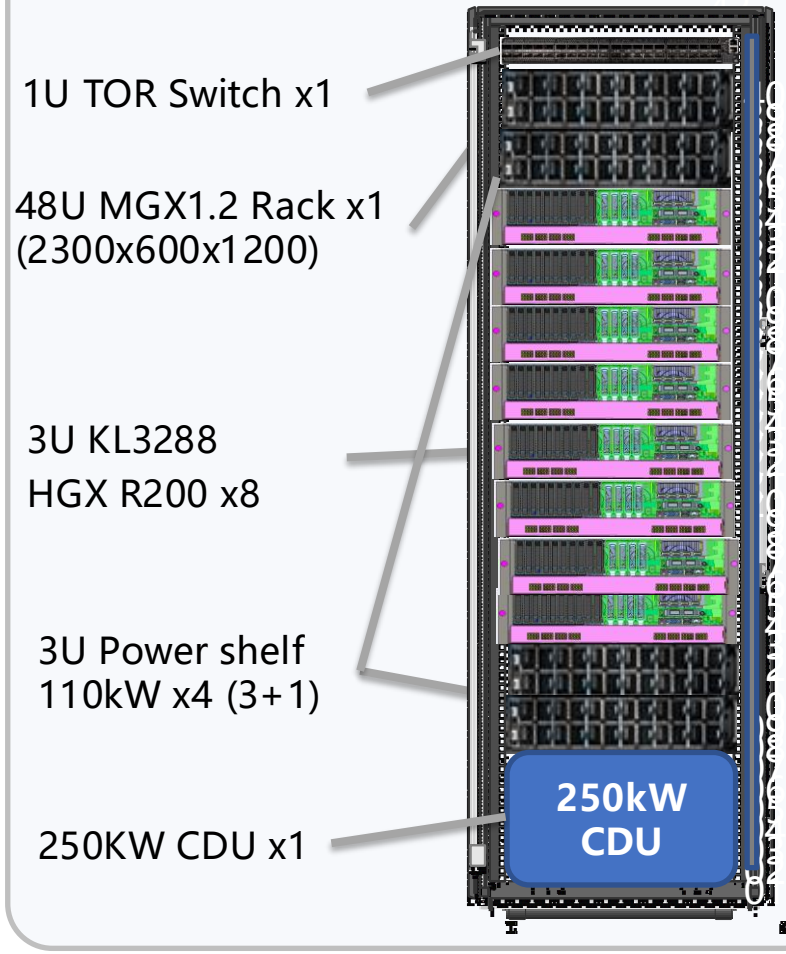
Power consumption is est. TDP 96 kW / EDP 101kVA per rack for 4 nodes

9 servers per rack (GPU density aligned with VR NVL72)



Power consumption is est. TDP 216 kW / EDP 227kVA per rack for 9 nodes

8 servers per rack (3U Hybrid-Liquid Rack)



Power consumption is est. TDP 192 kW / EDP 202kVA per rack for 8 nodes

Confidential – Subject to NDA

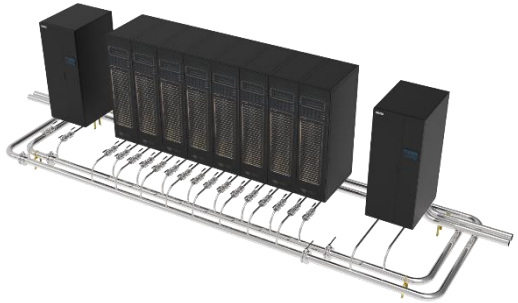
Aivres CDU Solutions for AIDC Scale Out

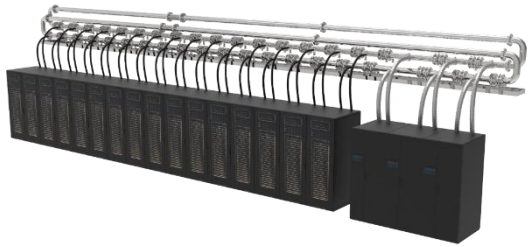




CU150A5

Comparison Items	In-Rack CDU	In-Row CDU
Delivery solution	In rack	Separate
Supported Quantity	1 rack	8+ racks
Redundancy Solution	Pump redundancy	CDU + pump redundancy
Secondary Loop	Not required	Required
Construction Period (8 x 16 Rack Pod)	Several days	1-2 months
Application Scenario	Small projects, fast delivery projects	Large-scale high-availability new data centers
Cost Advantage Range	Less than 8 racks	More than 16 racks
Advantage Summaryu	Fast delivery	High availability

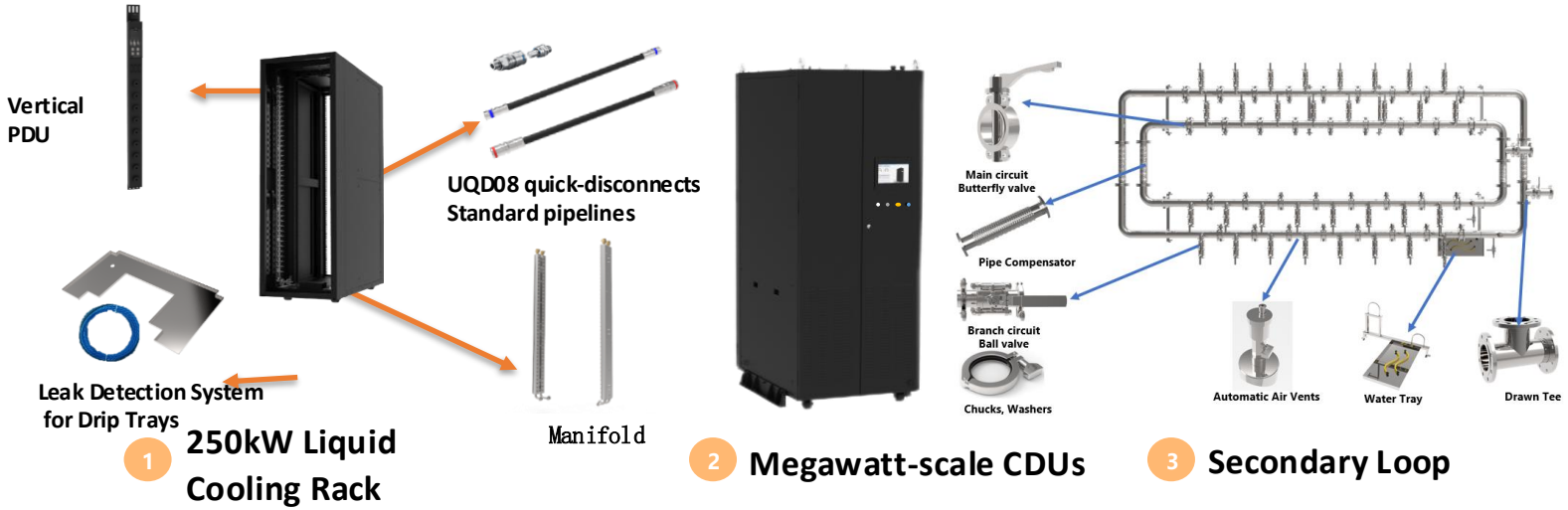






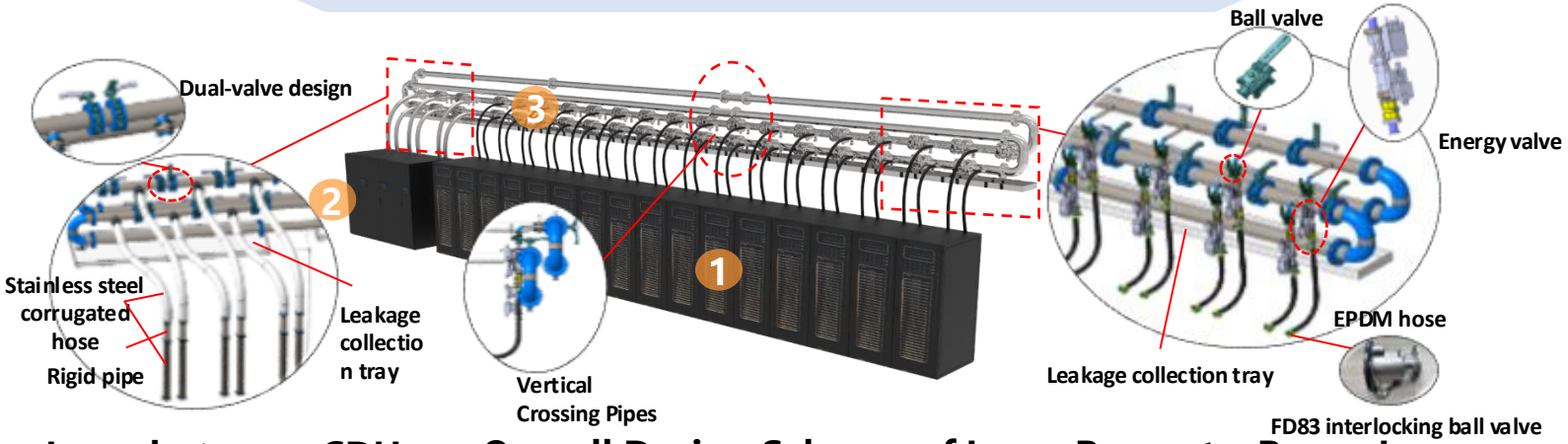
CU1200A4

AiVRES Liquid Cooling AIDC Solutions with KL2288/3288



Full-Stack Liquid Cooling Solution

- 250kW Liquid Cooling Rack
- Standardized liquid cooling components
- Smart Leak Detection System



Deep Customization for User Needs

- **Solution design：** Offers end-to-end customization service and adjusts key parameters based on actual scenarios.
- **Full-project BIM & 3D modeling**

700+KR5288V3+B200 Liquid-Cooled AIDC in Europe

- **Customer Request**
  - Cutomer 20MW DC in Europe, 4000+㎡ total ~100 Racks
  - L12 full-stack solution & turn-key delivery on-site

- **AiVRES Solution**
  - **GPU:** 6144 NVIDIA B200 GPUs 13824 PFLOP/s
  - **POD:** 3x POD(32xRack/POD+3xCDU with 2+1)
  - **Secondary Loop:** Secondary Loop with CDUs, Valves, Circuits, Coolant
  - **Implementation Services:** Installation and Integration for Pre-rack and Secondary Loop
  - **Maintenance Service:** 5 Years NBD Onsite Services and Routine Inspection Services



Deliver a modular liquid-cooled data center in 60 days

Summary

- AIDC architecture and challenges
- Innovative LC AIDC solutions and product lines
- Aivres DLC KRS8000V4/ KL2288/ KL3288 AI servers and solutions for LC AIDCs



Thank You

Our Booth #B3

To learn more, visit www.Aivres.com